Zachary Preator

Data Scientist | Machine Learning Engineer

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📍 Location: West Point, Utah

SUMMARY

Data Scientist and Machine Learning Engineer with 5+ years of experience spanning reliability engineering and production ML. Reliability-team background includes industrial statistics and design of experiments (JMP), attendance at the RAMS (Reliability and Maintainability Symposium) conference, and hands-on predictive modeling (XGBoost) on time-series sensor data. Mechanical engineering foundation brings physics-based reasoning to modeling problems, complementing production experience building cloud-based data pipelines (Apache Spark, AWS Glue) and deploying ML models at scale (AWS EKS/Kubernetes).

EXPERIENCE

Cricut, Machine Learning Engineer, South Jordan, Utah - *August 2023 - Present*

- Built large-scale data pipelines with Apache Spark in AWS Glue, feeding production model training and evaluation
- Fine-tuned CLIP on real and synthetic data for semantic image search; validated model performance and deployed to millions of users via AWS EKS/Kubernetes
- Architected and developed production APIs integrating LLMs into backend services
- Established model versioning, CI/CD, and deployment workflows using Docker, GitHub Actions, and ArgoCD
- Built an internal model-evaluation platform (Streamlit) for interactive analysis and stakeholder demos

Autoliv, Reliability Engineer, Ogden, Utah - *August 2020 - August 2023*

- Applied industrial statistics and design of experiments to manufacturing and test data; attended the RAMS (Reliability and Maintainability Symposium) conference
- Built XGBoost time-series models on tabular sensor data to predict physical test curves, applying PCA/dimensionality reduction and feature engineering
- Designed a semi-automated data annotation pipeline (YOLO) for manufacturing imagery, addressing limited and sparse labeled data through custom manual-refinement tooling
- Built and validated a CNN-based critical point detection model (TensorFlow), reducing analysis time by 80%
- Shipped a full-stack time-series anomaly detection system (Flask, SQLite, Plotly) applying statistical modeling to flag anomalies from thousands of real-time test deployments

Design Automation Associates Inc, Engineering Intern, Rexburg, Idaho - *May 2019 - June 2020*

- Built C# tooling automating workflows via CAD APIs; developed AR Android app for 3D assembly visualization

EDUCATION

Colorado State University - Global, Aurora, Colorado - *June 2021 - June 2024*

- Master of Science in Computer Science - AI & Machine Learning • GPA: 3.84

Brigham Young University - Idaho, Rexburg, Idaho - *August 2017 - December 2020*

- Bachelor of Science in Mechanical Engineering — Emphasis in Programming • GPA: 3.78

PROJECTS

Runalysis, *2026*

- Personal running-form analyzer applying YOLO pose estimation and point tracking to treadmill footage, extracting time-series biomechanical signals (cadence, stride length, ground contact time) and classifying form with an interpretable Random Forest model
- Technologies: Python, YOLO, OpenCV, NumPy

TECHNICAL SKILLS

Statistics & Reliability: Design of Experiments (JMP), PCA / Dimensionality Reduction, XGBoost, Reliability Engineering, Time-Series Modeling
Programming: Python, SQL, C#, JavaScript
ML/AI: PyTorch, TensorFlow, scikit-learn, CLIP, Hugging Face, YOLO, CNNs, Random Forest
Data & Cloud Engineering: Apache Spark, AWS Glue, AWS EKS, PostgreSQL, Redshift, Docker, Kubernetes, ArgoCD, Helm, GitHub Actions